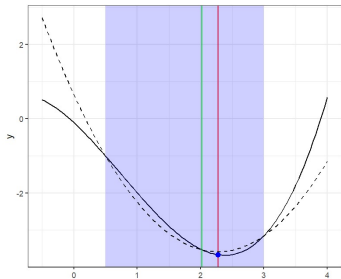
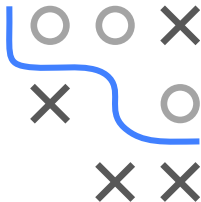


Optimization in Machine Learning

Univariate optimization

Brent's method



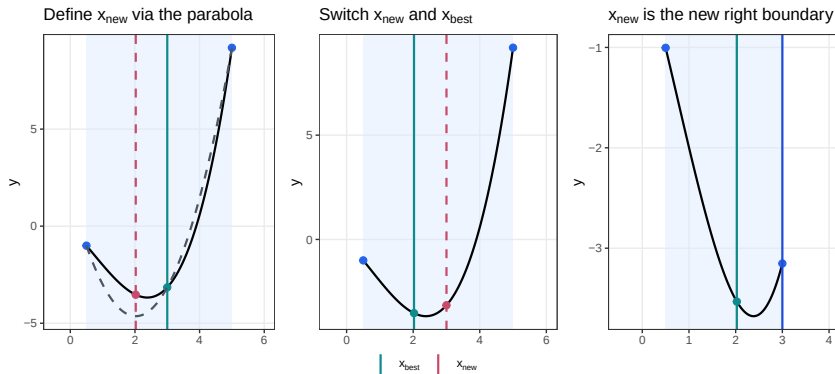
Learning goals

- Quadratic interpolation
- Brent's procedure

Slides based on [Press et al. 1992](#) (Ch. 10.2)

A 3x3 grid with a blue path starting at the top-left cell (0,0) and ending at the bottom-right cell (2,2). The path consists of the following cells: (0,0), (0,1), (1,1), (1,2), and (2,2). The cells (0,1), (0,2), (1,0), and (2,0) are blocked by grey 'X' marks. The cells (1,0), (2,1), and (2,2) are blocked by grey 'O' marks.

Similar to golden section search but select x_{new} differently:
minimum of parabola through $(x_{\text{left}}, f_{\text{left}}), (x_{\text{best}}, f_{\text{best}}), (x_{\text{right}}, f_{\text{right}})$

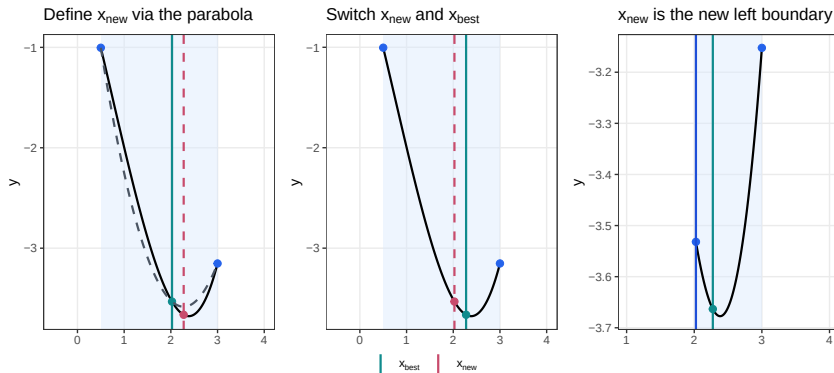
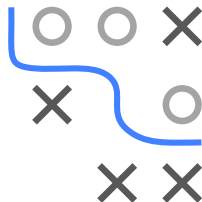


Left: Fit parabola (dashed) and propose its minimum (red) as new point.

Middle: Switch x_{new} and x_{best} if $f(x_{\text{new}}) < f_{\text{best}}$. Right: New interval.

QUADRATIC INTERPOLATION

Similar to golden section search but select x_{new} differently:
minimum of parabola through $(x_{\text{left}}, f_{\text{left}})$, $(x_{\text{best}}, f_{\text{best}})$, $(x_{\text{right}}, f_{\text{right}})$

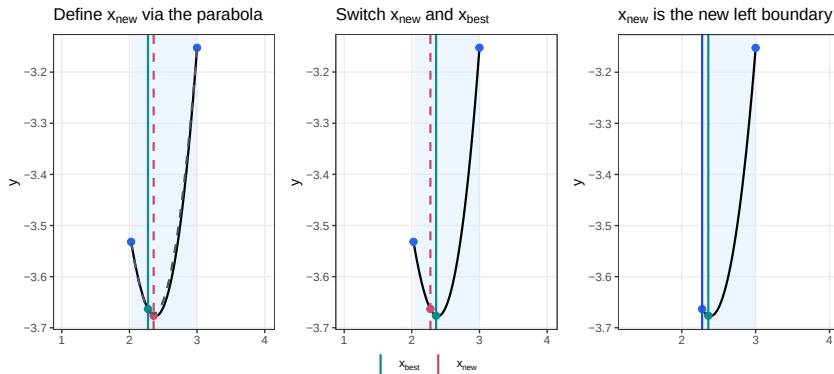
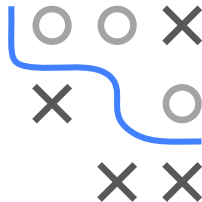


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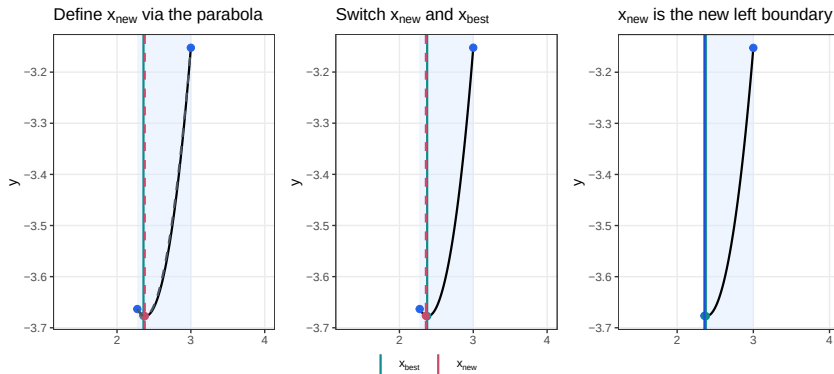
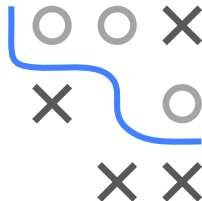


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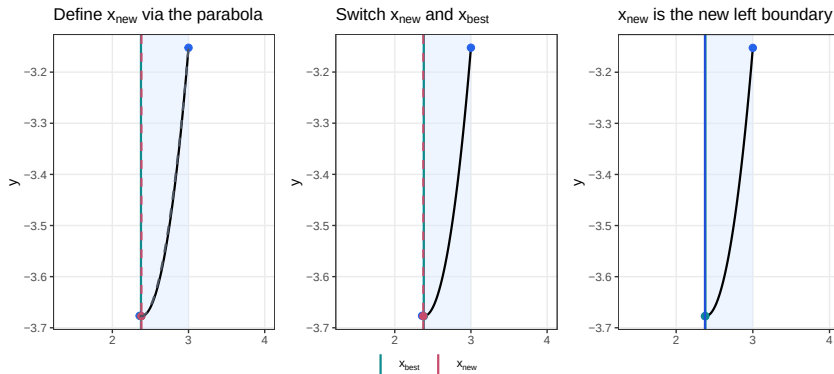


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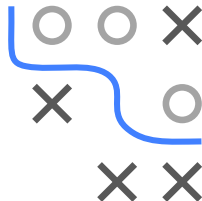
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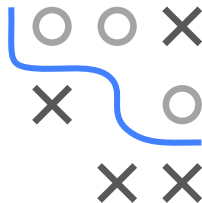
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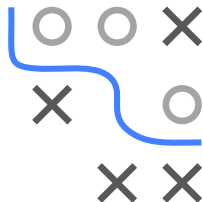
QUADRATIC INTERPOLATION COMMENTS

- Much faster than GSS for smooth f : well-approximated by a parabola near the minimum $\Rightarrow$ QI jumps directly toward the minimum
- But: not as robust as GSS, the following may happen:
 - Algorithm suggests very similar x_{new} in each step
 - x_{new} outside of search interval
 - Parabola degenerates to line and no real minimum exists
- Algorithm must then abort, finding the minimum is not guaranteed



BRENT'S METHOD

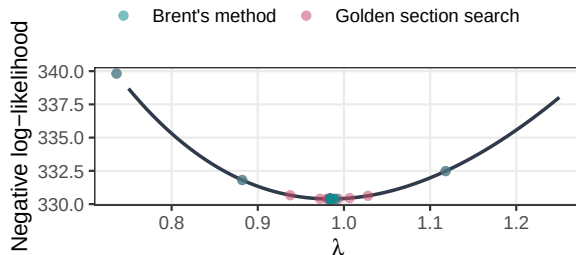
- Brent's algorithm ► Brent 1973 alternates between golden section search and quadratic interpolation as follows:
 - Quadratic interpolation step acceptable if:
 - (i) x_{new} falls within $[x_{\text{left}}, x_{\text{right}}]$
 - (ii) x_{new} sufficiently far away from x_{best}
(Heuristic: Less than half of the movement of step before last)
 - Otherwise: proposal via GSS
- GSS stabilizes unstable steps and guarantees **at least linear convergence** for any unimodal f
- Faster convergence (quadratic interpolation) on well-behaved (smooth) f : **superlinear convergence** of order ≈ 1.325
- Note: Brent's algorithm also exists with other robust methods than GSS, e.g., bisection method



EXAMPLE: MLE POISSON

- Poisson PMF: $p(k|\lambda) := \mathbb{P}(x = k) = \frac{\lambda^k \cdot \exp(-\lambda)}{k!}$
- Negative log-likelihood for n samples:

$$-\ell(\lambda, \mathcal{D}) = -\log \prod_{i=1}^n p(x^{(i)}|\lambda) = -\sum_{i=1}^n \log p(x^{(i)}|\lambda)$$



GSS and Brent converge to global min at $x^* \approx 1$

But GSS needs ≈ 45 iters, Brent only ≈ 15 for same tolerance

